

(a) $\frac{80q}{\epsilon_0}$ (b) $\frac{q}{40\epsilon_0}$

(c) $\frac{q}{80\epsilon_0}$ (d) $\frac{q}{160\epsilon_0}$

7. Two identical conducting balls having positive charges q_1 and q_2 are separated by a centre to centre distance r. If they are made to touch each other and then separated to the same distance, the force between them will be
(NSEP 04-05)

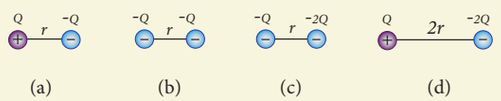
(a) less than before

(b) same as before

(c) more than before

(d) zero

8. Rank the electrostatic potential energies for the given system of charges in increasing order.



(a) 1 = 4 < 2 < 3 (b) 2 = 4 < 3 < 1

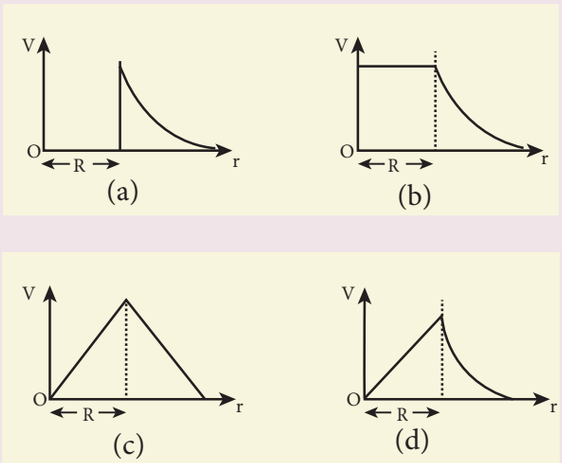
(c) 2 = 3 < 1 < 4 (d) 3 < 1 < 2 < 4

9. An electric field $\vec{E} = 10x\hat{i}$ exists in a certain region of space. Then the potential difference $V = V_o - V_A$, where V_o is the potential at the origin and V_A is the potential at x = 2 m is:

(a) 10 V (b) –20 V

(c) +20 V (d) –10 V

10. A thin conducting spherical shell of radius R has a charge Q which is uniformly distributed on its surface. The correct plot for electrostatic potential due to this spherical shell is



11. Two points A and B are maintained at a potential of 7 V and -4 V respectively. The work done in moving 50 electrons from A to B is

(a) 8.80×10^{-17} J

(b) -8.80×10^{-17} J

(c) 4.40×10^{-17} J

(d) 5.80×10^{-17} J

12. If voltage applied on a capacitor is increased from V to $2V$, choose the correct conclusion.

(a) Q remains the same, C is doubled

(b) Q is doubled, C doubled

(c) C remains same, Q doubled

(d) Both Q and C remain same

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